

MOS

CH

MOS

+ GAIN in $\textcircled{X}^4$ $\textcircled{1}$
CCND3/DAXX/LTB/
NOTCH4/PIN1

P1

TP53 17% $\longrightarrow$ 79%
RUNX1 2% $\longrightarrow$ 28%
SUZ12 2.4% $\longrightarrow$ 19%

EPKAS 6%
+ KDM6A 5.5%

P2

TET2 2.6% $\longrightarrow$ 37%
SRSF2 0.4% $\longrightarrow$ 21.6%

+ ASXL1
32.5%

P3

TET2 5% $\longrightarrow$ TET2 36% + TET2 2° hit 33%

P4

DNMT3A 9.8% $\longrightarrow$ 43%
TET2 22% $\longrightarrow$ 44%
TP53 5.5% $\longrightarrow$ 53.7%

+ GAIN $\textcircled{X}^2$
BRD4/
KAP1/
NOTCH3/
SPRINGER/
TYK2/VAV1

DNMT3A 0.5%
DNMT3A 0.3%
KLF4 0.5%

+ LOSS
CRLF2/P2RY8
 $\textcircled{X}^3$

ETERO GENETIC ?

P5

SPOP 0.8% $\longrightarrow$ 21%
TP53 1.5% $\longrightarrow$ 7%

PPH1D 0.7%
TP53 2° hit 0.3%
V2AF1 < 0.1%

P6

TET2 2° 0.3%
TET2 3° 0.3%
ZRSR2 0.2% $\longrightarrow$ 34%

(P7)

CM RDS ②
CHEK2 <0.1% → 8.7% + CHEK2 2° hit 7.5%
NCOR2 13.7%

(P8)

PPM1D <0.1%
AKT1 33.6% PRKAK1A 38.4%
CBL 2° 18.9% TET2 31.6%
CBL 2° 25.5%
+ WSS AURKA / PLCG1 / PTPN11 ④

(P9)

CHEK2 0.2% → 5.6%
DNMT3A 0.03% → 31.5%
TET2 0.24% → 9%
SARHDL / SRC / TOP1

(P10)

TET2 2.2% → 6.8%

(P11)

SF3B1 <0.1% → 2.4%
DIS3 1.4% → 13.9%

(P12)

TP53 1.3% → 16.1%

PPM1D 0.1%
DNMT3A 0.4%

(P13)

DNMT3A 0.3% → 5%

P14

CH

MOS

3

TET2 1° 0.1% → 41%
TET2 2° 0.04% → 38% + ETNK1 33.4%
SRSF2 0.3% → 41%
DNMT3A 1.2% → 41% + LOSS CUX1⁵
+
SH2B3 1° < 0.1%
SH2B3 2° < 0.1%

P15

CHK2 0.3% → 4.9%

P16

MSK6 < 0.1% → 8%
GNAS < 0.1% → 6.4%

P17

AXL 0.03% → 5.3%
DNMT3A < 0.01% → 3%

P2 → PROSTATE 73 ANM
P3 → LUNG 72 ANM
P6 → PROSTATE 72 ANM
P7 → BREAST 75 ANM
P9 → BREAST 74 ANM

P15 → CUTANEOUS 72 ANM
P17 → PANCREAS 57 ANM

P11 → SMALL CELL LUNG 83 ANM
P12 → UTERINE NA
P14 → OVARIAN 70 ANM

(MPN)

CK

~~TP53~~ MPN

(4)

(P18)

TP53 5.4% → TP53 87.5% + KRAS 23%

(P19)

ASXL1 0.7% → 7.6%
SAPK2 0.4% → 10%
+ LOSS MAPK3 ~~6~~

(P20)

→ HCC ASSOCIATO (64 ANNI)
DNMT3A 2% → 6.6%
CBL 0.6% → 3.2%

(A) #5 pazienti

CONCLUSIONE

~~TP53~~

(P4) | (P5) | (P6) | ~~(P7)~~

ABBIAMO CHE NELLA CHIP

CI SONO PIU' CLONI E SOLO

UN SOTTO INFERME DI QUESTI

POI ESPANDE NELL'MDS

→ IN #2 DI QUESTI PAZIENTI SI HANNO MUTATION

AGGIUNTE NELL'MDS RISPETTO A CHIP

#12 PAZIENTI MDS + IN TUTTI I 3 MPN

(P1) | (P2) | (P3) | (P7) | (P9) | (P10) | (P12) | (P13) | (P15) | (P16) | (P17)

NELLA CHIP CI SONO I CLONI CHE POI ESPANDONO

NELLA MDS/MPN. DI QUESTI #5 HANNO MUTATION
AGGIUNTE NELLA MDS E #1 NELLA MPN

(C) IN 1 CASO (P8) LA CH NON ESPANDE IN MDS
E LA MDS HA MOLTE MUTATION

N.B. → PPR10 IN CHIP È OBSERVATO #3 VOLTE MA NON
ESPANDE MAI IN MDS

→ IN ALCUNI CASI NON CI SONO NELLA MDS/MPN
MUTATION AD ALTA VAF

(P21) MYELODYSPLASTIC WORKUP

(5)

DNMT3A

CH

TUMOR

$< 0.01 \rightarrow 14\%$

(P22) THERAPY RELATED MDS

ATL 1° 0.2% $\rightarrow$ 35%

ATL 2° 0.3% $\rightarrow$ 23% + TP53 6%

N.B.

SETUP
OVARIAN
NEUA
CHIP

- TP53 IN CH CHE POI ESPANZE IN #5 VOLTE
- CHEK2 # 3 VOLTE
- ATL # 1 VOLTE (NEUA THERAPY RELATED)

DNMT3A # 6 VOLTE

TET2 # 6 VOLTE

(P23) ALL

CH

TUMOR

TP53 0.3% $\rightarrow$ 57%

CCNE1 $< 0.1\%$ $\rightarrow$ 19%

+ JAK1 20%
KMT2D 23%

N.B.

P21 $\rightarrow$ OVARIAN 63 ANNI

P22 $\rightarrow$ BLADDER 81 ANNI

P23 $\rightarrow$ OVARIAN 55 ANNI